



AIC, MATHEMATICS LEARNING AREA
YEAR 11 MATHEMATICS APPLICATIONS – UNIT 2

Assessment type: Response

TASK 9 – TEST 6

Applications of trigonometry

MARKING KEY

TIME ALLOWED FOR THIS PAPER

Reading and Working time for this paper: 50 minutes in class under test conditions

MATERIAL REQUIRED FOR THIS PAPER

TO BE PROVIDED BY THE SUPERVISOR

Question/answer booklet.

TO BE PROVIDED BY THE CANDIDATE

Standard Items: pens, pencils, pencil sharpener, highlighter, eraser, ruler, drawing templates

IMPORTANT NOTE TO CANDIDATES

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be attempted	Suggested working time (minutes)	Marks available
Calculator Assumed	7	7	50	42
			Marks available:	42
			Task Weighting	7%

Instructions to candidates

- The rules for the conduct of this examination are detailed in the booklet *WACE Examinations Handbook*. Sitting this examination implies that you agree to abide by these rules.
- Answer the questions in the spaces provided.
- Spare answer pages can be used. If you need to use them, indicate in the original answer space where the answer is continued.

SCSA Content – Topic 2.2: Applications of trigonometry

- 2.2.1 Use trigonometric ratios to determine the length of an unknown side, or the size of an unknown angle in a right-angled triangle
- 2.2.2 Determine the area of a triangle, given two sides and an included angle by using the rule $area = \frac{1}{2} ab \sin C$, or given three sides by using Heron's rule, and solve related practical problems
- 2.2.3 Solve problems involving non-right-angled triangles using the sine rule (acute triangles only when determining the size of an angle) and the cosine rule
- 2.2.4 Solve practical problems involving right-angled and non-right-angled triangles, including problems involving angles of elevation and depression and the use of bearings in navigation

**TO BE AWARDED FULL MARKS ALL WORKING OUT AND
CALCULATIONS MUST BE SHOWN**

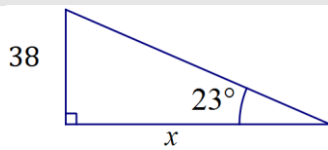
Question 1**(4 marks)**

(a) $\tan 60^\circ = \frac{5}{x} \rightarrow x = \frac{5}{\tan 60^\circ}$ ✓

$\therefore x = 2.89 \text{ m}$ ✓

(b) $\tan x^\circ = \frac{5}{8} \rightarrow x^\circ = \tan^{-1}\left(\frac{5}{8}\right)$ ✓

$\therefore x = 32.01^\circ$ ✓

Question 2**(3 marks)****Solution**

$$\tan 23 = \frac{38}{x} \Rightarrow x = 89.5 \text{ m}$$

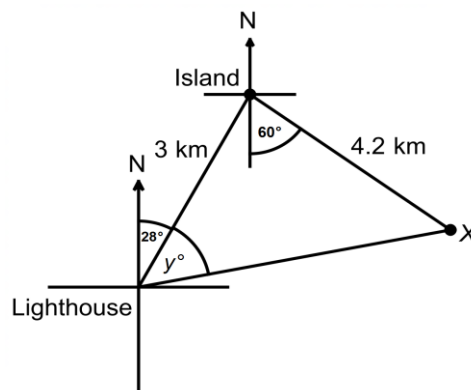
Specific behaviours

- ✓ sketch diagram with measurements
- ✓ uses tangent ratio
- ✓ states distance

Question 3**(9 marks)**

(a) 208°T ✓✓

(b)



(c) $x^2 = 3^2 + 4.2^2 - 2(3)(4.2)\cos 88^\circ$ ✓✓

$\therefore x = 5.075 \text{ km}$ ✓✓

(d) $\frac{\sin y}{4.2} = \frac{\sin 88}{5.075}$ ✓

$\therefore y = 55.8^\circ$ ✓

$\therefore \text{Bearing is } 56 + 28 = 084^\circ\text{T}$ ✓

Question 4**(7 marks)**

$$(a) \quad AB^2 = 35^2 + 42^2 - 2(35)(42)(\cos 120)$$

$$= 66.776$$

✓✓

$$\therefore 66.8\text{km}$$

✓

$$(b) \quad \frac{\sin x}{42} = \frac{\sin 120}{66.8}$$

$$x = 33^\circ$$

✓✓

$$\therefore \text{Bearing} = 360 - (33 + 145) = 182^\circ\text{T}$$

$$(c) \quad A = \frac{35 \times 42 \times \sin 120}{2}$$

$$= 636.5287$$

✓✓

[7]

$$\therefore 637\text{km}^2$$

Question 5**(7 marks)**

$$(a) \quad \sin 51 = \frac{6.32}{\text{gw1}}$$

✓✓

$$\text{Length of guywire 1} = 8.13\text{m} \quad \therefore 8\text{m}$$

$$(b) \quad \tan \theta = \frac{9.8}{6.32}$$

$$\theta = 57.18^\circ$$

✓✓

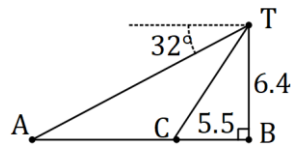
[4]

$$\therefore \text{Angle of depression} = 57^\circ$$

Question 6**(8 marks)**

- (a) Sketch a diagram to show the above information.

(2 marks)

Solution**Specific behaviours**

- ✓ vertical TB with angle of depression to A
- ✓ lengths TB and BC , making a right-angle

- (b) Calculate the line of sight distance from the top of the pole to A .

(2 marks)

Solution

$$\begin{aligned} AT &= 6.4 \div \sin 32 \\ &= 12.1 \text{ m} \end{aligned}$$

- (c) Determine the ang

(2 marks)

Specific behaviours

- ✓ uses sine ratio
- ✓ correct distance

$$= 49.3^\circ$$

Specific behaviours

- ✓ uses tangent ratio
- ✓ correct angle (to nearest degree)

- (d) Calculate the distance travelled by the animal from A to C .

(2 marks)

Solution

$$AB = 6.4 \div \tan 32 = 10.2 \text{ m}$$

$$AC = 10.2 - 5.5 = 4.7 \text{ m}$$

Specific behaviours

- ✓ calculates AB
- ✓ calculates AC

Question 7**(7 marks)**

(a) $PN = \sqrt{20^2 + 20^2 - 2(20)(20) \cos 44}$ ✓✓
= 14.98
∴ 15 metres

(b) $\frac{\sin OPN}{26} = \frac{\sin 30}{15}$ ✓
 $\angle OPN = 60^\circ$

$\angle MPN = 68^\circ$ since isosceles ✓

∴ $\angle MPN + \angle OPN$ ✓
 $68 + 60 = 128^\circ$

(c) $PO = \sqrt{15^2 + 26^2}$ ✓✓
= 30 metres